Liming Liu

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RESEARCH INTERESTS

Learning and Evaluating Reliable Adaptive Agents; Agent Capability Learning and Composition; Agent Reliability and Failure Diagnosis; Synthetic Environments for Agent Learning; Agentic Software and Enterprise Systems

EDUCATION

University of British Columbia Sep 2017 – May 2020

Master of Engineering in Electrical and Computer Engineering · Vancouver, Canada

Nanjing University of Aeronautics and Astronautics Sep 2013 – Jun 2017

Bachelor of Engineering in Information Engineering · China | Exchange: KAIST, South Korea (Sep–Dec 2015)

INDEPENDENT RESEARCH

Independent Researcher | Reliable Agentic Software Systems 2026 – Present

Where Does Software-Agent Delegation Fail?— *Independent Research (Github)*

- Designed and conducted a paired study across 30 SWE-Gate tasks (90 Planner runs and 270 Developer runs), comparing direct requirements, Planner-generated handoffs, and handoffs supplemented with original requirements.
- Estimated a 24.6% equal-task-weight rate of Direct-success/Delegated-failure outcomes across 23 eligible tasks. Conducted a prospective 270-run follow-up comparing source anchoring with identical-prompt retries, finding no established average advantage for restoring original requirements.
- Performed outcome-blinded failure analysis and evaluator audits to distinguish requirement-transmission failures from downstream execution failures, examine alternative explanations, and assess measurement validity.

Verified Procedural Skill Composition for Small Language-Model Agents — *Independent Research (Github)*

- Built a deterministic, offline Jira–GitHub workflow simulator covering ticket creation, sprint planning, sprint rollover, and pull-request synchronization, with state-based validation of execution outcomes and operational constraints.
- Fine-tuned SmolLM2-360M Base and Instruct models on atomic workflow decisions across three training seeds each, establishing per-category and closed-loop proficiency criteria that all six models met at epoch 4.
- Evaluated three qualified Base-model checkpoints on 16 shared, unseen workflow-composition scenarios. None completed a scenario, despite producing 510/510 structurally valid actions, motivating further investigation of task and interface generalization.
- Conducted controlled comparisons of task labels, state representations, and tool availability. Found that expanding the displayed and permitted tool interface disrupted previously correct atomic rollover decisions; redesigned the follow-up study to establish shared-interface proficiency before testing composition.

PROFESSIONAL EXPERIENCE

Full Stack Developer Jan 2022 – May 2026

SAP · Vancouver, Canada

- **SAP Partner Finder — AI Search & Retrieval:** Designed and implemented a RAG-style partner-recommendation PoC for 3,000+ global SAP partners using Python, Qdrant, embeddings, hybrid retrieval, and reciprocal rank fusion. Developed a country-aware representation strategy for structured and unstructured partner profiles to improve semantic retrieval across multi-country service data.
- **SAP Partner Exception Management — GenAI Workflow Assistance:** Built a GenAI PoC that translated approvers' compliance criteria into structured prompts and evaluation logic, tested the approach against

historical rejected submissions, and generated potential rejection risks and revision suggestions before submission.

- **Enterprise Product Engineering:** Led the UI redesign implementation of SAP Partner Finder in Vue.js and contributed to Spring Boot/HANA search and partner-contact services. The globally deployed application reached 97% user satisfaction
- **Distributed-System Performance:** Reduced Partner Finder search latency from approximately 5 seconds to 300 ms by identifying repeated cross-system data fetches as the bottleneck, validating the source data's low update frequency, and replacing real-time retrieval with a 24-hour synchronization and caching strategy.
- **Reliability, Security & Observability:** Improved production reliability through Jenkins/Honeycomb CI/CD observability, Kibana/OpenSearch logging and request tracing, automated testing, and remediation of CSRF and XSS vulnerabilities identified during pre-launch penetration testing.

Full Stack Developer, iXp Intern Sep 2020 – Dec 2021

SAP · Vancouver, Canada

- Developed core search and partner-contact functionality for SAP Partner Finder using Vue.js and Spring Boot, contributing across both UI and backend services.
- Contributed to the application's UI redesign implementation preceding its global launch and continued on the project as a full-time developer after the internship.

RESEARCH TEAM EXPERIENCE

Centre for Heart Lung Innovation | Software Engineer Intern May 2019 – Dec 2019

Vancouver, Canada

- Worked in a cross-disciplinary computational genomics team with medical researchers and biostatisticians, translating analysis and data-management requirements into reliable research workflows.
- Developed data upload, download, and bulk-update capabilities for a human-genome (GRCh38) management system; collaborated on SLURM, Bash, AWK, and R workflows for preliminary statistical analysis.
- Migrated a legacy Struts application to Spring Boot and built unit and end-to-end tests with JUnit and Selenium to reduce operational risk in active research pipelines.

CERTIFICATIONS

- SAP Certified – SAP Generative AI Developer
- SAP Certified – Solution Architect – SAP BTP
- Claude Certified Architect – Foundations
- Claude Certified Developer – Foundations

TECHNICAL SKILLS

AI & GenAI: SAP BTP AI Core, SAP GenAI Hub (OpenAI/ChatGPT API via BTP), SAP Translation Hub, Vector Databases (Qdrant), RAG, Prompt Engineering, LLM Integration, Python

Backend: Java, SpringBoot, Node.js, SAP CAP, RESTful APIs, OData, Microservices, Hibernate, JPA

Frontend: SAP UI5, SAP Fiori, Vue.js, React.js, JavaScript, HTML/CSS

Cloud & DevOps: SAP BTP (AI Core, Destination Service, Object Store Service), Jenkins (CI/CD), Docker, SAP HANA, MySQL

Analytics & Data: SAP Web Analytics, Adobe Analytics, Qualtrics, Bash, AWK, R, MATLAB

Testing: JUnit, Nightwatch.js, Cucumber, Selenium